# Problem # 1

The op amp in the circuit is ideal.

a) Calculate $v_o$ if $v_a = 4\text{V}$ and $v_b = 0\text{V}$.

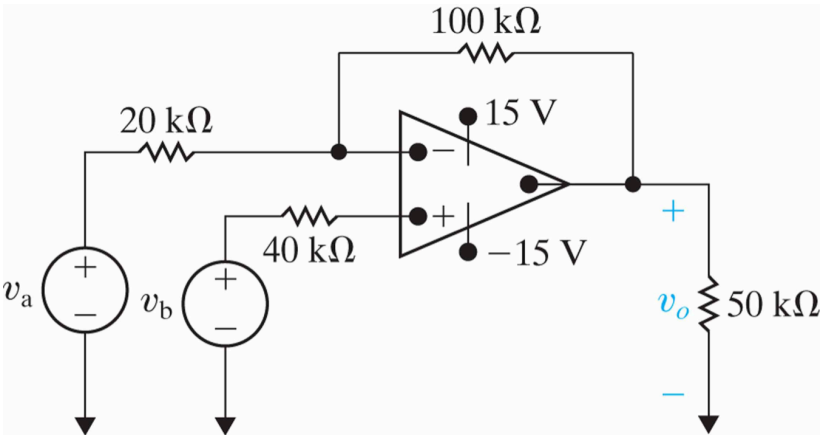
b) Calculate $v_o$ if $v_a = 2\text{V}$ and $v_b = 0\text{V}$.

c) Calculate $v_o$ if $v_a = 2\text{V}$ and $v_b = 1\text{V}$.

d) Calculate $v_o$ if $v_a = 1\text{V}$ and $v_b = 2\text{V}$.

e) Calculate $v_o$ if $v_a = 1.5\text{V}$ and $v_b = 4\text{V}$.

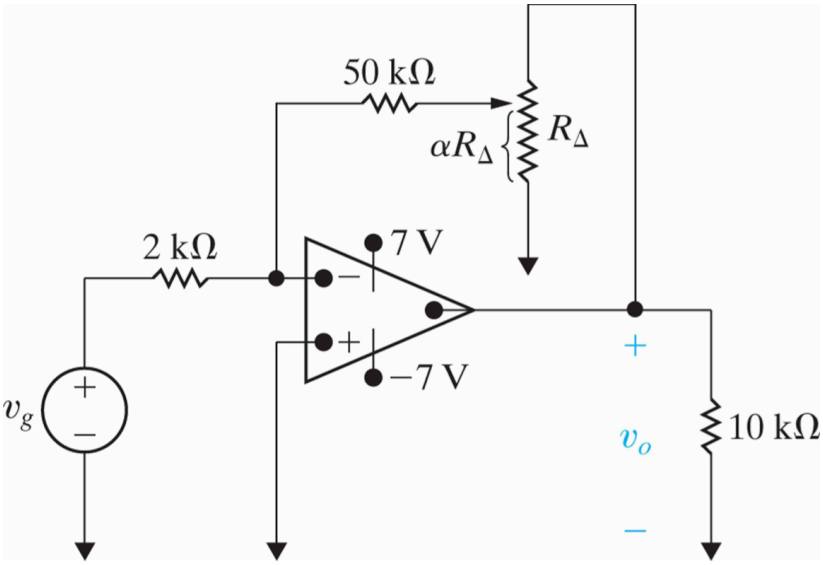
If $v_b = 1.6$ V, specify the range of $v_a$ such that the amplifier does not saturate.



# Problem #2

a) The op amp in the circuit shown is ideal. The adjustable resistor $R_\Delta$ has a maximum value of 100 kΩ, and $\alpha$ is restricted to the range of $0.2 < \alpha < 1$. Calculate the range of $v_o$ if $v_g = 40$ mV.

b) If $\alpha$ is not restricted, at what value of $\alpha$ will the op amp saturate?




# Problem #3

Design an inverting-summing amplifier so that

$$v_O = -(3\ v_a + 5\ v_b + 4\ v_c + 2\ v_d).$$

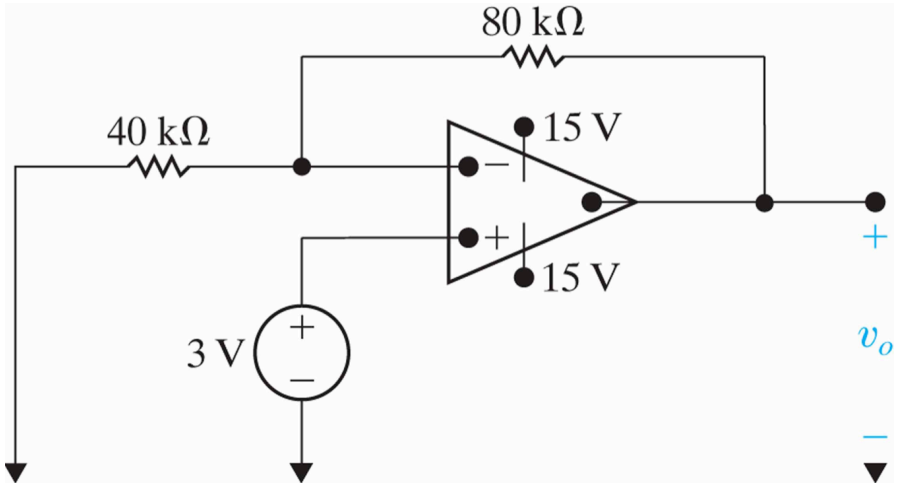
Start by choosing a feedback resistor ($R_f$) from Appendix H. Then choose single resistors from Appendix H or construct resistor networks from resistors in Appendix H to satisfy the design values for $R_a$, $R_b$, $R_c$, and $R_d$. Draw your final circuit diagram.

# Problem #4

The op amp in the circuit is ideal.

a) What op amp circuit configuration is this?

b) Calculate $v_o$.




# Problem #5

The op amp in the circuit is ideal. What value of $R_f$ will give the equation $v_o = 5 - 4v_a$, for this circuit?

